# Semeton Balogun

Backend and systems engineer, moving into solutions architecture

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## SUMMARY

Backend and systems engineer with six years in production: multi-tenant platforms for banks and enterprise SaaS, offline-first data for national health campaigns, and governance for AI coding agents. I start from the business goal: sizing infrastructure, designing deployments inside bank networks, and representing engineering with stakeholders. My current research is on agent governance: guardrails and repository memory for coding agents.

## EXPERIENCE

**eGov Foundation** — Hub Software Engineer, West Africa (Contract) Jun 2026 – present

Bridge between country programme teams and platform engineering on DIGIT HCM, the open-source platform running Nigeria's national malaria campaigns.

- Diagnosed activity dates being stamped at sync time instead of on-device creation time, which locked field workers out of working days and skewed attendance-based pay; documented the root cause and fix requirement
- Contributed to the stock-reconciliation design for an ITN distribution pilot in an offline-first ledger
- Sizing infrastructure and support staffing for a 2027–29 programme across 11 states (~27M target population)

**Seamless Technologies** — Backend Engineer Nov 2024

Recruitment platform and HRMS modules for a multi-tenant HR suite serving enterprise clients.

- Built the Kafka pipeline to the company data lake as a non-blocking side effect, so a mapping failure can't fail the primary write
- Implemented the two-way integration between the recruitment platform and the core HRMS, including SSO with company-access checks and service-to-service auth
- Led tenant-isolation and security hardening: SSRF-safe webhooks, company-scoped reads and writes, per-client mail credentials, GDPR self-service deletion
- Backend for AI-assisted hiring: CV parsing, candidate ranking, automated screening
- Volunteer facilitator on the Codeable programme, teaching software engineering to Special Olympics athletes and students at the Kirikiri correctional centre

**Zercom Systems** — Technical Lead Apr 2023 – Nov 2024

Led the rebuild of a per-client LMS into Tangerine365, one multi-tenant platform running on-premise at banks, as a self-serve cloud product, and inside Human Manager HRMS.

- Merged per-client codebases into one vanilla PHP 8 codebase with tenancy by subdomain or by ID, so a fix ships once instead of once per client
- Acted as solutions architect for client deployments: architecture, server sizing and specifications, and engineering's voice in stakeholder meetings
- Designed First Bank's three-tier on-premise deployment; deployed across IIS inside bank networks, Apache, DigitalOcean and AWS
- Wrote a SQL Server driver alongside the MySQL default, chosen at runtime, for banks that required it
- Consolidated the company's internal tools, staging and UAT onto one CyberPanel/LiteSpeed server with wildcard subdomains, cutting that hosting cost ~85%

**ScholarshipIQ** — Backend Engineer Jun 2022 – Apr 2023

- Cut API response times ~50% through query optimisation and indexing; introduced staging and branching discipline, cutting production errors by over 90%

**Bincom Dev Center** — Backend Developer (apprenticeship) Sep 2020 – Jul 2022

- Backend work in Node.js and PHP on ARM Learn for ARM Pensions and Social Lender; trained 5+ interns

## SELECTED PROJECTS AND RESEARCH

**Lawyering**
Workspace for law firms and individual practitioners, built agent-first as a modular monolith of 19 Laravel modules with tenant scoping, encrypted file sharing and billing in naira and US dollars. In production since June 2026, with over 1,200 automated tests.

**Repository memory for coding agents**
Talk for PHP Talks Lagos and an experiment harness measuring whether a verified, human-gated decision log changes agent behaviour, including memory poisoning.

**TrustCenta**
Multi-tenant security platform wrapping OWASP ZAP, Gophish and CISO Assistant; rebuilt after a compromise and wrote the public post-mortem.

**Agent Contract**
npm CLI that checks every write an AI coding agent attempts against repository rules and blocks out-of-scope changes before they land.

**Terminoxx360**
ATM fleet monitoring at Access Bank; sole engineer on a multi-tenant rebuild with CBN regulatory uptime, automated escalation and an on-premise AI option.

## EDUCATION

**Miva Open University** — Master of Information Technology, specialising in AI — In progress

**University of Lagos** — BSc Nursing Science, Distinction (joint-highest CGPA) — 2015 – 2021
Registered Nurse, Nursing and Midwifery Council of Nigeria, 2022.

## SKILLS

**Architecture:** multi-tenancy, deployment design and infrastructure sizing, queues and event-driven workflows, offline-first sync, security hardening

**Languages and frameworks:** PHP (Laravel and vanilla), TypeScript, Node.js, React, Next.js, Livewire

**Data and infrastructure:** MySQL, PostgreSQL, SQL Server, Redis, Kafka, Elasticsearch, Docker, Linux, IIS, Apache, LiteSpeed, DigitalOcean, AWS

**AI:** Claude, OpenAI and Ollama integrations; agent-first development; governance for coding agents

## WRITING

Engineering essays on Codeplified (semeton.substack.com), including *The Architecture I Didn't Design* and *Beyond Prompts: Repository Governance for AI Coding Agents*.